LIF overview

ONTAP 9

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LIF overview

Configure LIFs overview

A LIF (logical interface) represents a network access point to a node in the cluster. You can configure LIFs on ports over which the cluster sends and receives communications over the network.

A cluster administrator can create, view, modify, migrate, revert, or delete LIFs. An SVM administrator can only view the LIFs associated with the SVM.

A LIF is an IP address or WWPN with associated characteristics, such as a service policy, a home port, a home node, a list of ports to fail over to, and a firewall policy. You can configure LIFs on ports over which the cluster sends and receives communications over the network.

> Beginning with ONTAP 9.10.1, firewall policies are deprecated and wholly replaced with LIF service policies. For more information, see Configure firewall policies for LIFs.

LIFs can be hosted on the following ports:

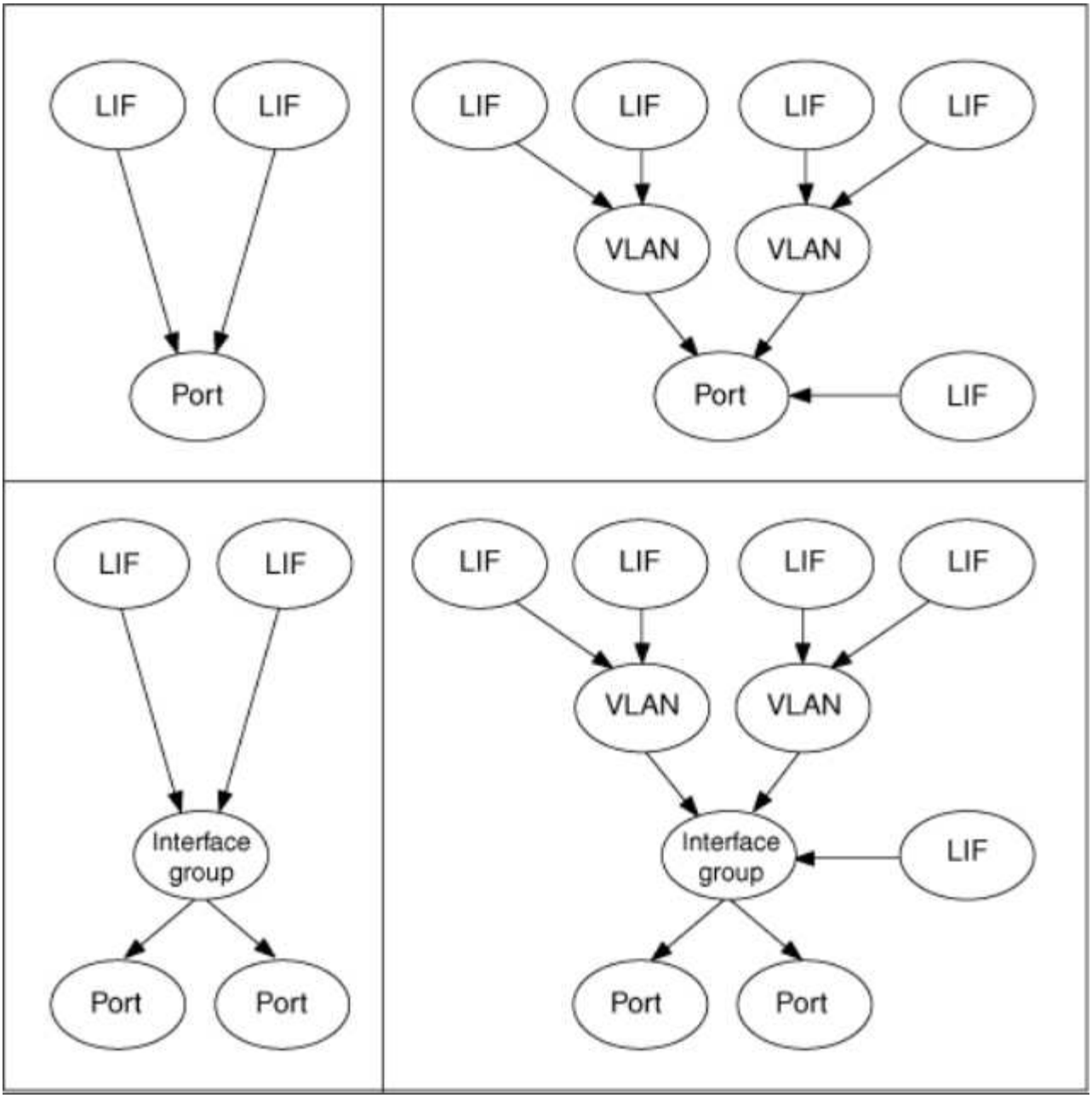
- Physical ports that are not part of interface groups
- Interface groups
- VLANs
- Physical ports or interface groups that host VLANs
- Virtual IP (VIP) ports

  Beginning with ONTAP 9.5, VIP LIFs are supported and are hosted on VIP ports.

While configuring SAN protocols such as FC on a LIF, it will be associated with a WWPN.

SAN administration

The following figure illustrates the port hierarchy in an ONTAP system:



LIF failover and giveback

A LIF failover occurs when a LIF moves from its home node or port to its HA partner node or port. A LIF failover can be triggered automatically by ONTAP or manually by a cluster administrator for certain events such as a down physical Ethernet link or a node dropping out of replicated database (RDB) quorum. When a LIF failover occurs, ONTAP continues normal operation on the partner node until the reason for the failover is resolved. When the home node or port regains health, the LIF is reverted from the HA partner back to its home node or port. This reversion is called a giveback.

For LIF failover and giveback, ports from each node need to belong to the same broadcast domain. To check that the relevant ports on each node belong to the same broadcast domain, see the following:

- ONTAP 9.8 and later: Repair port reachability
- ONTAP 9.7 and earlier: Add or remove ports from a broadcast domain

For LIFs with LIF failover enabled (either automatically or manually), the following applies:

- For LIFs using a data service policy, you can check failover-policy restrictions:
  - ONTAP 9.6 and later: LIFs and service policies in ONTAP 9.6 and later
  - ONTAP 9.5 and earlier: LIF roles in ONTAP 9.5 and earlier
- Auto-revert of LIFs happens when the auto-revert is set to `true` and when the LIF's home port is healthy and able to host the LIF.
- On a planned or unplanned node takeover, the LIF on the node that is taken over, fails over to the HA partner. The port on which the LIF fails over is determined by VIF Manager.
- After the failover is complete, the LIF operates normally.
- When a giveback is initiated, the LIF reverts back to its home node and port, if auto-revert is set to `true`.
- When an ethernet link goes down on a port hosting one or more LIFs, the VIF Manager migrates the LIFs from the down port to a different port in the same broadcast domain. The new port could be in the same node or its HA partner. After the link is restored and if auto-revert is set to `true`, the VIF Manager reverts the LIFs back to their home node and home port.
- When a node drops out of replicated database (RDB) quorum, the VIF Manager migrates the LIFs from the out of quorum node to its HA partner. After the node comes back into quorum and if auto-revert is set to `true`, the VIF Manager reverts the LIFs back to their home node and home port.

LIF compatibility with port types

LIFs can have different characteristics to support different port types.

> When intercluster and management LIFs are configured in the same subnet, the management traffic might be blocked by an external firewall and the AutoSupport and NTP connections might fail. You can recover the system by running the `network interface modify -vserver` *`vserver name`* `-lif` *`intercluster LIF`* `-status-admin up|down` command to toggle the intercluster LIF. However, you should set the intercluster LIF and management LIF in different subnets to avoid this issue.

LIF	Description
Data LIF	A LIF that is associated with a storage virtual machine (SVM) and is used for communicating with clients. You can have multiple data LIFs on a port. These interfaces can migrate or fail over throughout the cluster. You can modify a data LIF to serve as an SVM management LIF by modifying its firewall policy to mgmt. Sessions established to NIS, LDAP, Active Directory, WINS, and DNS servers use data LIFs.
Cluster LIF	A LIF that is used to carry intracluster traffic between nodes in a cluster. Cluster LIFs must always be created on cluster ports. Cluster LIFs can fail over between cluster ports on the same node, but they cannot be migrated or failed over to a remote node. When a new node joins a cluster, IP addresses are generated automatically. However, if you want to assign IP addresses manually to the cluster LIFs, you must ensure that the new IP addresses are in the same subnet range as the existing cluster LIFs.

Cluster management LIF	LIF that provides a single management interface for the entire cluster. A cluster management LIF can fail over to any node in the cluster. It cannot fail over to cluster or intercluster ports
Intercluster LIF	A LIF that is used for cross-cluster communication, backup, and replication. You must create an intercluster LIF on each node in the cluster before a cluster peering relationship can be established. These LIFs can only fail over to ports in the same node. They cannot be migrated or failed over to another node in the cluster.
Node management LIF	A LIF that provides a dedicated IP address for managing a particular node in a cluster. Node management LIFs are created at the time of creating or joining the cluster. These LIFs are used for system maintenance, for example, when a node becomes inaccessible from the cluster.
VIP LIF	A VIP LIF is any data LIF created on a VIP port. To learn more, see Configure virtual IP (VIP) LIFs.

LIFs and service policies (ONTAP 9.6 and later)

You can assign service policies (instead of LIF roles or firewall policies) to LIFs that determine the kind of traffic that is supported for the LIFs. Service policies define a collection of network services supported by a LIF. ONTAP provides a set of built-in service policies that can be associated with a LIF.

You can display service policies and their details using the following command:

```
network interface service-policy show
```

Features that are not bound to a specific service will use a system-defined behavior to select LIFs for outbound connections.

Service policies for system SVMs

The admin SVM and any system SVM contain service policies that can be used for LIFs in that SVM, including management and intercluster LIFs. These policies are automatically created by the system when an IPspace is created.

The following table lists the built-in policies for LIFs in system SVMs as of ONTAP 9.12.1. For other releases, display the service policies and their details using the following command:

```
network interface service-policy show
```

Policy	Included services	Equivalent role	Description
default-intercluster	intercluster-core, management-https	intercluster	Used by LIFs carrying intercluster traffic. Note: Service intercluster-core is available from ONTAP 9.5 with the name net-intercluster service policy.

default-route-announce	management-bgp	-	Used by LIFs carrying BGP peer connections Note: Available from ONTAP 9.5 with the name net-route-announce service policy.
default-management	management-core, management-https, management-http, management-ssh, management-autosupport, management-ems, management-dns-client, management-ad-client, management-ldap-client, management-nis-client, management-ntp-client, management-log-forwarding	node-mgmt, or cluster-mgmt	Use this system scoped management policy to create node- and cluster-scoped management LIFs owned by a system SVM. These LIFs can be used for outbound connections to DNS, AD, LDAP, or NIS servers as well as some additional connections to support applications that run on behalf of the entire system. Beginning in ONTAP 9.12.1, you can use the `management-log-forwarding` service to control which LIFs are used to forward audit logs to a remote syslog server.

The following table lists the services that LIFs can use on a system SVM as of ONTAP 9.11.1:

Service	Failover limitations	Description
intercluster-core	home-node-only	Core intercluster services
management-core	-	Core management services
management-ssh	-	Services for SSH management access
management-http	-	Services for HTTP management access
management-https	-	Services for HTTPS management access
management-autosupport	-	Services related to posting AutoSupport payloads
management-bgp	home-port-only	Services related to BGP peer interactions
backup-ndmp-control	-	Services for NDMP backup controls
management-ems	-	Services for management messaging access
management-ntp-client	-	Introduced in ONTAP 9.10.1. Services for NTP client access.

management-ntp-server	-	Introduced in ONTAP 9.11.1. Services for NTP server management access
management-portmap	-	Services for portmap management
management-rsh-server	-	Services for rsh server management
management-snmp-server	-	Services for SNMP server management
management-telnet-server	-	Services for telnet server management
management-log-forwarding	-	Introduced in ONTAP 9.12.1. Services for audit log forwarding

Service policies for data SVMs

All data SVMs contain service policies that can be used by LIFs in that SVM.

The following table lists the built-in policies for LIFs in data SVMs as of ONTAP 9.11.1. For other releases, display the service policies and their details using the following command:

```
network interface service-policy show
```

Policy	Included services	Equivalent data protocol	Description
default-management	management-https, management-http, management-ssh, management-dns-client, management-ad-client, management-ldap-client, management-nis-client	none	Use this SVM-scoped management policy to create SVM management LIFs owned by a data SVM. These LIFs can be used to provide SSH or HTTPS access to SVM administrators. When necessary, these LIFs can be used for outbound connections to an external DNS, AD, LDAP, or NIS servers.
default-data-blocks	data-core, data-iscsi	iscsi	Used by LIFs carrying block-oriented SAN data traffic. Starting in ONTAP 9.10.1, the "default-data-blocks" policy is deprecated. Use the "default-data-iscsi" service policy instead.

default-data-files	data-fpolicy-client, data-dns-server, data-flexcache, data-cifs, data-nfs, management-dns-client, management-ad-client, management-ldap-client, management-nis-client	nfs, cifs, fcache	Use the default-data-files policy to create NAS LIFs supporting file-based data protocols. Sometimes there is only one LIF present in the SVM, therefore this policy allows the LIF to be used for outbound connections to an external DNS, AD, LDAP, or NIS server. You can remove these services to from this policy if you prefer these connections utilize only management LIFs.
default-data-iscsi	data-core, data-iscsi	iscsi	Used by LIFs carrying iSCSI data traffic.
default-data-nvme-tcp	data-core, data-nvme-tcp	nvme-tcp	Used by LIFs carrying NVMe/TCP data traffic.

The following table lists the services that can be used on a data SVM along with any restrictions each service imposes on a LIF's failover policy as of ONTAP 9.11.1:

Service	Failover restrictions	Description
management-ssh	-	Services for SSH management access
management-http	-	Introduced in ONTAP 9.10.1 Services for HTTP management access
management-https	-	Services for HTTPS management access
management-portmap	-	Services for portmap management access
management-snmp-server	-	Introduced in ONTAP 9.10.1 Services for SNMP server management access
data-core	-	Core data services
data-nfs	-	NFS data service
data-cifs	-	CIFS data service
data-flexcache	-	FlexCache data service
data-iscsi	home-port-only	iSCSI data service
backup-ndmp-control	-	Introduced in ONTAP 9.10.1 Backup NDMP controls data service

data-dns-server	-	Introduced in ONTAP 9.10.1 DNS server data service
data-fpolicy-client	-	File-screening policy data service
data-nvme-tcp	home-port-only	Introduced in ONTAP 9.10.1 NVMe TCP data service
data-s3-server	-	Simple Storage Service (S3) server data service

You should be aware of how the service policies are assigned to the LIFs in data SVMs:

- If a data SVM is created with a list of data services, the built-in "default-data-files" and "default-data-blocks" service policies in that SVM are created using the specified services.
- If a data SVM is created without specifying a list of data services, the built-in "default-data-files" and "default-data-blocks" service policies in that SVM are created using a default list of data services.

  The default data services list includes the iSCSI, NFS, NVMe, SMB, and FlexCache services.

- When a LIF is created with a list of data protocols, a service policy equivalent to the specified data protocols is assigned to the LIF.
- If an equivalent service policy does not exist, a custom service policy is created.
- When a LIF is created without a service policy or list of data protocols, the default-data-files service policy is assigned to the LIF by default.

Data-core service

The data-core service allows components that previously used LIFs with the data role to work as expected on clusters that have been upgraded to manage LIFs using service policies instead of LIF roles (which are deprecated in ONTAP 9.6).

Specifying data-core as a service does not open any ports in the firewall, but the service should be included in any service policy in a data SVM. For example, the default-data-files service policy contains the following services by default:

- data-core
- data-nfs
- data-cifs
- data-flexcache

The data-core service should be included in the policy to ensure all applications using the LIF work as expected, but the other three services can be removed, if desired.

Client-side LIF service

Beginning with ONTAP 9.10.1, ONTAP provides client-side LIF services for multiple applications. These services provide control over which LIFs are used for outbound connections on behalf of each application.

The following new services give administrators control over which LIFs are used as source addresses for

certain applications.

Service	SVM restrictions	Description
management-ad-client	-	Beginning with ONTAP 9.11.1, ONTAP provides Active Directory client service for outbound connections to an external AD server.
management-dns-client	-	Beginning with ONTAP 9.11.1, ONTAP provides DNS client service for outbound connections to an external DNS server.
management-ldap-client	-	Beginning with ONTAP 9.11.1, ONTAP provides LDAP client service for outbound connections to an external LDAP server.
management-nis-client	-	Beginning with ONTAP 9.11.1, ONTAP provides NIS client service for outbound connections to an external NIS server.
management-ntp-client	system-only	Beginning with ONTAP 9.10.1, ONTAP provides NTP client service for outbound connections to an external NTP server.
data-fpolicy-client	data-only	Beginning with ONTAP 9.8, ONTAP provides client service for outbound FPolicy connections.

Each of the new services are automatically included in some of the built-in service policies, but administrators can remove them from the built-in policies or add them to custom policies to control which LIFs are used for outbound connections on behalf of each application.

LIF roles (ONTAP 9.5 and earlier)

LIFs with different roles have different characteristics. A LIF role determines the kind of traffic that is supported over the interface, along with the failover rules that apply, the firewall restrictions that are in place, the security, the load balancing, and the routing behavior for each LIF. A LIF can have any one of the following roles: cluster, cluster management, data, intercluster, node management, and undef (undefined). The undef role is used for BGP LIFs.

Beginning with ONTAP 9.6, LIF roles are deprecated. You should specify service policies for LIFs instead of a role. It is not necessary to specify a LIF role when creating a LIF with a service policy.

LIF security

	Data LIF	Cluster LIF	Node management LIF	Cluster management LIF	Intercluster LIF

Require private IP subnet?	No	Yes	No	No	No
Require secure network?	No	Yes	No	No	Yes
Default firewall policy	Very restrictive	Completely open	Medium	Medium	Very restrictive
Is firewall customizable?	Yes	No	Yes	Yes	Yes

LIF failover

	Data LIF	**Cluster LIF**	**Node management LIF**	**Cluster management LIF**	**Intercluster LIF**
Default behavior	Only those ports in the same failover group that are on the LIF's home node and on a non-SFO partner node	Only those ports in the same failover group that are on the LIF's home node	Only those ports in the same failover group that are on the LIF's home node	Any port in the same failover group	Only those ports in the same failover group that are on the LIF's home node
Is customizable?	Yes	No	Yes	Yes	Yes

LIF routing

	Data LIF	**Cluster LIF**	**Node management LIF**	**Cluster management LIF**	**Intercluster LIF**
When is a default route needed?	When clients or domain controller are on different IP subnet	Never	When any of the primary traffic types require access to a different IP subnet	When administrator is connecting from another IP subnet	When other intercluster LIFs are on a different IP subnet
When is a static route to a specific IP subnet needed?	Rare	Never	Rare	Rare	When nodes of another cluster have their intercluster LIFs in different IP subnets

When is a static host route to a specific server needed?	To have one of the traffic types listed under node management LIF, go through a data LIF rather than a node management LIF. This requires a corresponding firewall change.	Never	Rare	Rare	Rare

LIF rebalancing

	Data LIF	Cluster LIF	Node management LIF	Cluster management LIF	Intercluster LIF
DNS: use as DNS server?	Yes	No	No	No	No
DNS: export as zone?	Yes	No	No	No	No

LIF primary traffic types

	Data LIF	Cluster LIF	Node management LIF	Cluster management LIF	Intercluster LIF
Primary traffic types	NFS server, CIFS server, NIS client, Active Directory, LDAP, WINS, DNS client and server, iSCSI and FC server	Intracluster	SSH server, HTTPS server, NTP client, SNMP, AutoSupport client, DNS client, loading software updates	SSH server, HTTPS server	Cross-cluster replication

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